Distorted Vision: Testing the Robustness of X-ray Classification Models

1 Introduction

One of the most exciting applications of artificial intelligence (AI) in healthcare lies in radiology, with the diagnosis of lung conditions gaining much attention in the aftermath of Covid-19. Machine learning models can achieve good performance (approaching certified radiologist level) in standard, optimal conditions (Rahman et al., 2021). When models are exposed to real-world adverse conditions (e.g. X-ray distortions, overexposure), they often perform worse.

This report uses three different models to explore the research question, *"Testing the robustness and performance of X-ray classification models under real world conditions"*. It was initially predicted that the simpler models would perform poorly, while the complex models would handle most real-world conditions with ease. This report contains an outline of the methodology (models, preprocessing etc.), accompanied by results and critical analysis, followed by a concise summary of the main findings. A subset of approximately 10,000 X-ray images was selected from a publicly available dataset (see next section for details).

2 Literature Review

This report employs data from the Covid-19 Radiography set (Chowdhury et al., 2020, Rahman et al., 2021). It contains over 20,000 X-rays across four categories: Covid-19, Normal, Lung Opacity and Viral Pneumonica. Due to the limited compute available (1x Nvidia 3080 GPU), only a subset containing the first two categories was retained.

Recent work by Martínez-Martínez, 2025, highlights the crucial importance of robustness in medical imaging, indicating that deep learning models are often vulnerable to both adversarial attacks and other naturally occuring distortions, with work shown in the paper through use of data augmentation techniques to improve model robustness. This report complements the above research by ascertaining how robust models are under commonly occuring adverse conditions. A Vision Transformer (ViT) was also implemented, as presented by Dosovitskiy et al., 2022, at ICLR. While the lectures do (briefly) address transformer architectures, its scope is limited to sequence processing and neural machine translation - omitting its potential applications in image classification.

3 Methodology

3.1 Preprocessing

The X-rays were resized to 120x120 pixels (from 300x300), worsening model performance only slightly (∼2%), while dramatically increasing processing speed (x4). All X-rays were converted to grayscale - further reducing training time. When retaining only two classes, the dataset was fairly imbalanced (10,000 normal vs 3,600 Covid-19 X-rays), so 3,500 normal X-rays were removed to further reduce the class imbalance.

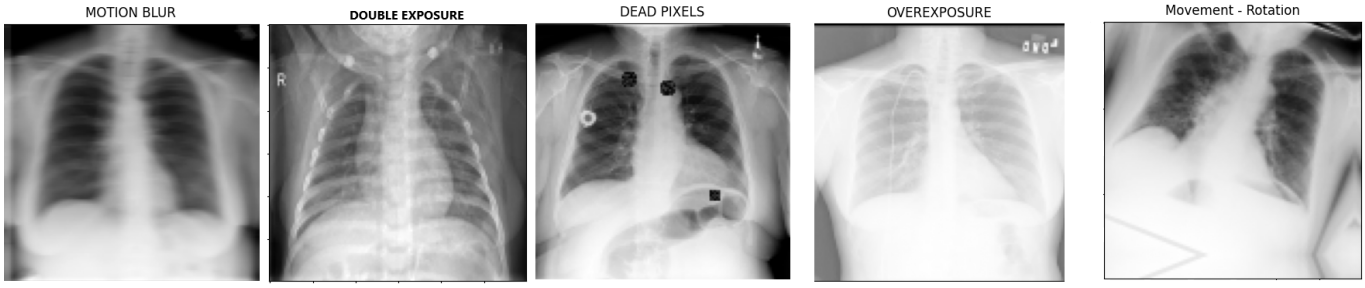

Figure 1: Examples of adverse real-world conditions simulated

Real-world adverse conditions were simulated (see figure 1) using image pre-processing, such as motion blur (patient didn't stand still), double exposure (X-ray receptor was not reset between scans), dead-pixel effects (detector malfunction), overexposure (X-ray beam too powerful), along with geometric distortions including rotation, translation, and projection (positioning errors in patient or radiation source).

3.2 Cross Validation and Hyperparameter Tuning

To ensure robust and consistent results, nested k-fold partitioning was used, with 10 outer and 3 inner folds for hyperparameter tuning (and evaluation), all implemented from scratch using a custom function wrapper, simplifying model implementation. Only the most salient hyperparameters tested are addressed in this report.

3.3 Model Development

Three models were chosen (of increasing complexity). K-Nearest Neighbours (KNN), Convolutional Neural Network (CNN) and a Vision Transformer (ViT). All three were implemented using PyTorch. For the CNN and ViT models, early stopping was enabled (patience 10, delta 0.02).

3.3.1 KNN (Baseline Model)

KNN is an instance-based, non parametric algorithm. Instances are classified based on the closest K instances in the training data (depending on the distance metric selected). The two hyperparameters were number of neighbours (1, 5, 40) and distance metric (Manhattan, Euclidian, Minkowski), all neighbours were weighted equally.

3.3.2 Convolutional Neural Network

Compared to a traditional neural network, the advantage of CNN lies with its ability to preserve spatial information and generally having less parameters due to use of convolutional kernels, also enabling the model to detect the same features at different places in an image. The two hyperparameters were number of layers (2, 4, 6) and kernel size (3, 5, 7). All models started with 7 filters (initial layer), with subsequent layers doubling the number of filters and each convolutional layer being followed by a MaxPooling layer. Two dense layers (sizes 312, 156) were placed prior to output layer. A maximum limit of 60 epochs was chosen.

3.3.3 Vision Transformer - ViT

Vision Transformers (ViT) take a novel approach to image classification (Dosovitskiy et al., 2022). Unlike CNNs, which rely on receptive fields, ViTs divide an image into fixed-size patches and process them as a sequence, which are linearly embedded prior being fed into a transformer encoder and enabling global attention across the entire image from the first layer. This approach was applied to model the relationships between different parts of an image. Compared to the others, this model requires more careful hyperparameter tuning due to its computational complexity and architectural sensitivity. As ViTs lack the strong inductive biases inherent in CNNs (e.g. built-in translation equivariance), they usually rely on large datasets and optimised regularisation to improve generalistion. A limit of 30 epochs was chosen. Two key hyperparameters were tested: initial learning rate (1e-05, 1e-03, 1e-02) and transformer depth (1, 6, 20).

4 Results and Discussion

Due to the sheer number of results generated, only the average results (including uncertainty, 5% CI) are reported for outer partitions in the nested cross validation (across the metrics).

4.1 KNN

KNN achieved high accuracy on normal X-rays (91.3%) but fell notably for more severe distortions (double exposure, over exposure, movement rotation). In particular, over-exposure (57.6%) had a decline of over 30%. Minor distortions (motion blur, dead pixels) generally had a smaller impact on results. The large perfomance difference between the less and more severe distortions can be attributed to the fact that KNN directly compares pixel values (to compute "distance"), leaving it quite sensitive to the more extreme distortion types (see table 1).

An interesting trend when comparing precision and recall is that for less adverse conditions and standard X-rays, precision is higher than recall (missing Covid instances but with fewer false positives), meaning the model is biased towards classifying instances as not having Covid, this can be explained by KNN's underlying nature. For imbalanced datasets (as in this case), it will be biased towards the majority class (i.e. not having Covid), as classification is based on a majority vote of the K nearest neighbours. Interestingly, for more severe conditions, the opposite occurs (higher recall than precision), falsely classifying relatively more instances as having Covid-19. This means that some non Covid instances after being transformed bear some resemblance to Covid-19 instances (at least at a pixel level), hence the misclassifications. Note also the low Kappa score for overexposure, indicating that there is only a small amount of agreement (in terms of identifying Covid-19 instances) beyond what would be expected by random chance.

Metric	**Normal**	**Motion Blur**	**Double Exposure**	**Dead Pixels**	**Over-exposure**	**Movement Rotation**
Accuracy	91.3%±0.6	90.4%±0.6	69.8%±1.3	91.3%±0.5	57.6%±1.3	70.7%±1.3
Recall	85.6%±1.7	84.5%±1.6	74.3%±2.1	85.5%±1.5	77.6%±3.8	85.5%±1.1
Precision	89.6%±1.7	88.1%±2.1	60.0%±1.6	89.9%±1.7	44.6%±1.5	55.9%±1.5
Specificity	94.4%±1.0	93.7%±0.8	67.4%±1.8	94.6%±1.1	46.4%±2.1	62.5%±1.8
F1-Score	87.5%±0.8	86.3%±0.9	63.8%±1.4	87.6%±0.7	56.6%±2.0	67.6%±1.2
Kappa-Score	80.8±1.3	78.9±1.5	38.8±0.8	80.9±1.1	20.6±2.6	42.9±2.1
Jaccard Score	77.8±1.3	75.9±1.5	46.9±1.5	77.9±1.1	39.5±1.9	51.1±1.4

Table 1: Overall results for KNN model

Regarding hyperparameters, Manhattan distance with K = 5 consistently yielded the best results across all partitions. Comparatively, using Euclidean distance reduced performance by about 3%, and Minkowski distance would worsen performance by around 6%. For K, using a single neighbour (K = 1) would worsen performance by around 2% (too sensitive to noise), while setting K = 40 would worsen (comparative) performance by approximately 3% (large K underfits). Overall, KNN proved quite robust at handling real-world conditions. Overall training time was 30 minutes.

4.2 CNN

The CNN model achieves an accuracy of 85.6% on standard X-rays (see table 2). When minor distortions such as motion blur or dead pixels were introduced, the model experienced moderate falls in accuracy (10%-15%). However,

strong distortions generally caused even larger drops in performance.

Metric	Normal	Motion Blur	Double Exposure	Dead Pixels	Over-exposure	Movement Rotation
Accuracy	85.6%±2.3	73.6%±2.1	71.6%±2.7	80.8%±3.8	53.5%±1.9	55.5%±2.3
Recall	79.0%±6.1	62.9%±6.7	58.6%±8.6	70.6%±7.7	90.0%±3.0	81.0%±2.1
Precision	80.4%±2.6	63.9%±3.3	61.0%±4.4	74.4%±4.5	42.9%±1.7	43.5%±1.9
Specificity	89.3%±1.5	80.1%±3.2	78.9%±3.7	86.5%±2.5	33.3%±2.4	41.2%±4.0
F1-Score	79.4%±3.8	62.9%±3.8	59.0%±5.2	72.2%±5.8	58.0%±1.8	56.6%±1.7
Kappa-Score	68.5±5.3	42.9±5.0	37.5±6.6	57.6±8.5	18.7±3.1	18.7±3.0
Jaccard Score	66.3±5.4	46.2±4.1	42.3±5.4	57.1±7.0	41.0±1.8	39.4±1.7

Table 2: Overall results for Convolutional Neural Network

Surprisingly, when comparing CNN performance to the previous model, KNN actually outperformed CNN, especially for motion blur and movement-rotation (also experienced less fall in performance). Comparable performance was observed for double and over-exposure. Another interesting trend to note is that the uncertainties are comparatively much larger (twice as large on average), especially for double exposure and dead pixel distortions (indicating high variability in CNN's performance and overfitting).

The cause of the poorer performance can be attributed to the fact that the the learning rate (from 5e-6 to 3e-4) was increased to ensure training time was within a reasonable amount (less than 3s per epoch) due to nested cross-validation. But this also appears to have caused the model to overshoot the optimal weights at times. The higher variability can also be attributed to the reason given above, a lower learning rate would have (almost certainly) led to more consistent results. Across all partitions, 2 convolutional layers performed best (deeper architectures led to increased overfitting). For kernel size, a size of five led to the best overall results. Using smaller (3) or larger (7) kernels would slightly reduce the accuracy (by ∼4% and ∼3% respectively). Using deeper layers (4 or 6) led to lower accuracy as well (3% and 10% on average). The overall training time (nested CV) was 4.5 hours.

4.3 ViT

Under normal conditions, ViT has an accuracy of 91.3% and an F1-score of 87.7%, indicating strong all-round performance. However, its performance reduces under certain distortion types, particularly over-exposure and movement rotation, where accuracy decreases to 68.4% and 58.9% respectively (see table 3).

Metric	Normal	Motion Blur	Double Exposure	Dead Pixels	Over-exposure	Movement Rotation
Accuracy	91.3%±0.5	88.6%±0.9	73.3%±3.1	90.8%±0.5	68.4%±3.6	58.9%±3.5
Recall	87.6%±1.8	84.5%±1.6	78.0%±5.8	87.4%±1.9	82.4%±7.4	86.7%±5.2
Precision	87.9%±1.5	83.9%±1.9	61.0%±4.7	87.1%±2.0	55.3%±4.8	46.6%±2.8
Specificity	93.3%±1.1	91.0%±0.9	70.8%±7.6	92.8%±1.2	60.7%±9.8	43.4%±8.3
F1-Score	87.7%±0.9	84.2%±1.5	67.7%±1.6	87.2%±0.9	65.2%±1.0	60.2%±1.0
Kappa-Score	80.9±1.1	75.3±2.1	45.9±4.1	80.1±1.3	38.9±4.3	25.4±3.9
Jaccard Score	78.1±1.4	72.7±2.2	51.2±1.7	77.3±1.5	48.4±1.1	43.1±1.1

Table 3: Overall results for ViT model

When comparing model performance, ViT outperforms CNN across virtually all performance metrics (and all transformation types), especially for motion blur and overexposure. This is no surprise as the ViT's (relatively) faster training time ensured that the model's performance was not downgraded to ensure a reasonable running time. Another factor (why ViT outperforms CNN), is likely due to vision transformers processing images as a sequence of patches and self self-attention mechanism, being able to understand the relationships between distant parts of an image, while CNN has an inductive bias towards local features due to its convolutional nature. Typically, in situations in which the dataset is relatively small (as is the case here), one would expect CNNs to perform better as their inductive bias permits them to learn more efficiently from a limited dataset (this did not occur due to performance downgrade). Vision transformers often require much more data to overcome the lack of inductive bias. But on huge datasets, the transformer architecture allows them to draw out complex relationships between features (Takahashi, 2024).

Moreover, ViT also has modest confidence intervals (indicating more stable results), being higher than KNN and lower than CNN. Across all partitions, the hyperparameter values that performed best were a learning rate of 1e-3 and a transformer depth of 6. Using a learning rate of 1e-5 (lower) or 1e-2 (higher) would worsen performance (on average) by 10% and 16% respectively. Changing the transformer depth to 1 (causing underfitting) or 20 (overfitting) would worsen typical performance by 4% and 2% respectively. The overall training time for ViT was 90 minutes.

4.4 Discussion

A decision tree was *briefly* considered as the baseline model (instead of KNN). Typical testing showed slightly worse performance (by approximately 4%). Moreover, the decision tree had higher variance in performance, indicating a degree of overfitting. However, the main reason why KNN was chosen was due to the overall running time being

shorter (7x faster). KNN's runtime is almost wholly due to inference (being a lazy learner). While a decision tree is more computationally expensive to train, it is quicker in terms of inference. For a problem that involved relatively less training (compared to the amount of inference), then a decision tree may have been the better choice. Regarding the dataset chosen, a more complex 14-class dataset was initially examined, with instances having one or more diseases. After conducting initial testing, which yielded poor results (∼40% average accuracy), it seemed logical to search for a simpler dataset, as the primary purpose of the research was to examine classifier robustness (not to simply compare disease prediction ability).

For the final dataset used, lung masks were provided (see figure 2), being used in a variety of lung imaging tasks. Naturally, it was thought that only using the lung data itself (after applying the mask) would improve the results (elimination of a lot of background "noise"). However, the opposite occurred (results fell by ∼15%). The likely cause is that automatically generated lung masks are often incorrect (leading to parts of the lung being cropped).

Based on a comprehensive evaluation, it appears that ViT is the model which is most robust to handling these (adverse) real-world conditions, especially when primarily considering accuracy (general overall performance) and recall (minimise false negatives). KNN performed surprisingly well (and had the lowest overall variance), in contrast to the initial prediction that KNN would perform worse, as it often struggles with high-dimensional data (due to sparse representation) and the model's underlying inability to learn features (which one would assume would be crucial to detecting Covid-19).

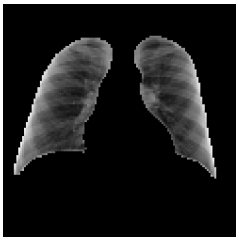

Figure 2: Example of an applied lung mask

4.5 Limitations

The main limitation experienced was the amount of compute available, resulting in many compromises being made, which affected the usefulness of the findings, in particular CNN, which experienced performance degradation due to resource constraints. Moreover, these included a reduction in the number of training instances, restricting the task to two classes and the number of hyperparameters tested. A further issue is the risk of bias in the dataset used, as it appears to be primarily sourced from Western Europe/USA, potentially reducing the usefulness of this report's findings for those located outside of these regions.

4.6 Future Extensions

With more resources, a future project could undertake testing with a larger/more diverse multi-class dataset, testing more complex models and a larger selection of hyperparameters (e.g. deeper model architectures). In addition, a wider variety of (more computationally expensive) real-world conditions could be modeled (e.g. simulate a patient forgetting to remove jewelry, static electricity effects).

5 Conclusion

Assisting radiologists with diagnosing lung diseases is an exciting application for machine learning. Problems often occur (due to various causes) when X-rays are captured. Under these sub-optimal conditions, machine learning models can often struggle to maintain the same level of performance compared to standard X-rays. This report analyses the performance of three machine learning models in detecting Covid-19 under real-world adverse conditions, simulated through preprocessing. Surprisingly, KNN achieved good performance, being almost equal to ViT, despite its simplistic nature. CNN performed unusually poorly due to resource limitations. ViT was the most robust, with decent overall accuracy and strong recall for the majority of adverse distortions - demonstrating the power of self-attention mechanisms, being able to capture complex relationships from across the image and being fairly resistant to distortions. Overall, the findings in this report provide practical guidance and insights for selecting and deploying robust machine learning systems in clinical imaging environments where data quality is an issue.

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